# L2L Framework: Data Preparation

LAPOP Mexico 2023 & Latinobarómetro Colombia 2023

Kevin Linares (JPSM, University of Maryland)

2026-02-24

```
pacman::p_load(sjlabelled, psych, knitr,
               viridis, patchwork, lavaan, semTools, tidyverse)

set.seed(721)

results_dir <- "D:/repos/UMD_classes_code/TSE_II_SURV721/results"
dir.create(results_dir, recursive = TRUE, showWarnings = FALSE)
```

## Overview

This script prepares two survey datasets for the Latent-to-Language (L2L) framework:

- **LAPOP Mexico 2023** — AmericasBarometer, 18-item trust/system support battery
- **Latinobarómetro Colombia 2023** — 9-item institutional trust battery

For each survey, we:

- (1) load raw data
- (2) inspect value labels to understand coding
- (3) select and recode variables
- (4) assess missingness and drop high-missing demographics
- (5) apply listwise deletion
- (6) validate the final dataset.

## LAPOP Mexico 2023

### Load and Inspect

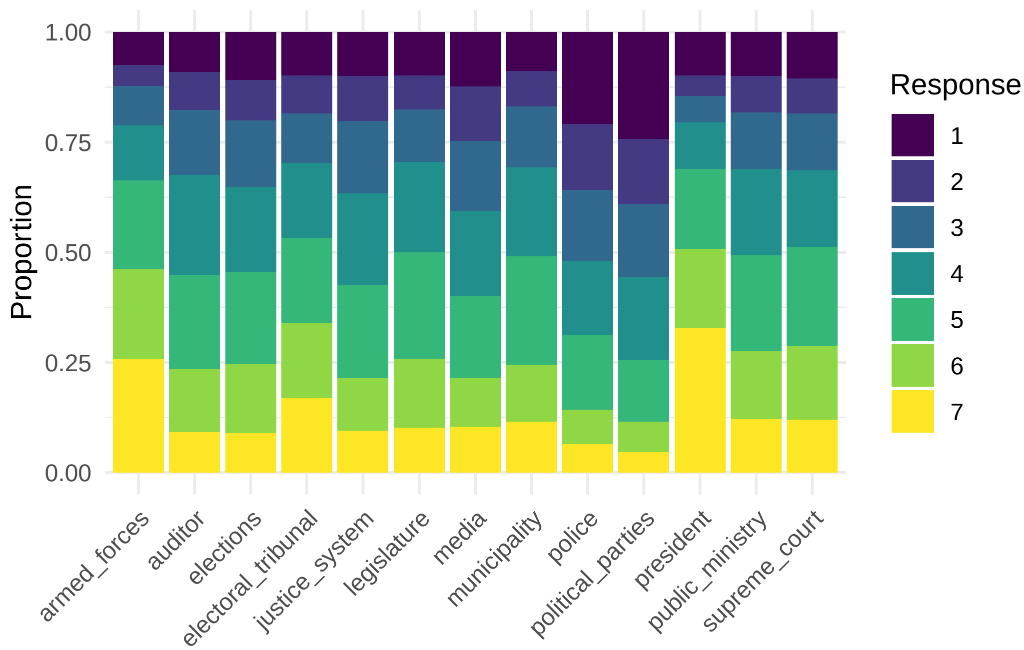
```
lapop_raw <- read_spss(
  "D:/repos/UMD_classes_code/TSE_II_SURV721/lapop/MEX_2023_LAPOP_AmericasBarometer_v1.0_w.sav") |>
  tibble()
```

```
Converting atomic to factors. Please wait...
```

#### Trust items "Trust in"

Coded as 1 = Not at all, . . ., 7 = a lot.

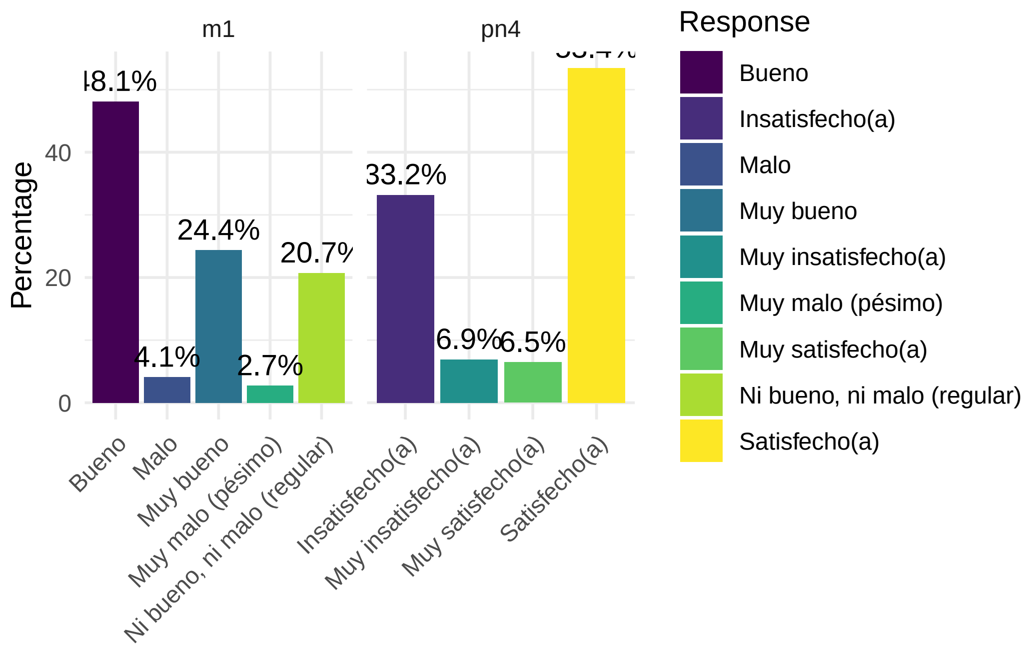
```r
trust_items <- c("b10a", "b11", "b12", "b13", "b15",
                 "b18", "b19", "b21", "b21a", "b31",
                 "b32", "b37", "b47a")


lapop_trust_names <- c(
  justice_system = "b10a",
  electoral_tribunal = "b11",
  armed_forces = "b12",
  legislature = "b13",
  public_ministry = "b15",
  police = "b18",
  auditor = "b19",
  political_parties = "b21",
  president = "b21a",
  supreme_court = "b31",
  municipality = "b32",
  media = "b37",
  elections = "b47a"
)


lapop_raw |>
  select(all_of(trust_items)) |>
  rename(!!!lapop_trust_names) |>
  drop_na() |>
  pivot_longer(everything(), names_to = "item", values_to = "value") |>
  count(item, value) |>
  ggplot(aes(x = item, y = n, fill = factor(value))) +
  geom_col(position = "fill") +
  scale_fill_viridis_d() +
  labs(x = NULL, y = "Proportion", fill = "Response") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```




Trust items

### Outcome Variables

```
lapop_raw |>
  select(pn4, m1) |>
  mutate_all(sjlabelled::as_character) |>
  pivot_longer(everything(),
               names_to = "Outcome", values_to = "Response") |>
  drop_na() |>
  group_by(Outcome, Response) |>
  count() |>
  group_by(Outcome) |>
  mutate(pct = n / sum(n) * 100) |>
  ggplot(aes(x = Response, y = pct, fill = Response)) +
  geom_col() +
  geom_text(aes(label = sprintf("%.1f%%", pct)), vjust = -0.5) +
  facet_wrap(~Outcome, scales = "free_x") +
  labs(y = "Percentage", x = NULL) +
  scale_fill_viridis_d() +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```




**M1 is presidential rating. The following are the labels translated from Spanish.**

- 1 = Very good
- 2 = Good
- 3 = Neighter good nor bad
- 4 = Bad
- 5 = Very Bad

**pn4 is satisfaction with democracy**

- 1 = Very satisfied
- 2 = satisfied
- 3 = dissatisfied
- 4 = very dissatisfied

## Recode data

```
# demographics
lapop_demo_source <- c("q1tc_r", "q2", "edre", "ur", "ocup4a")

lapop <- lapop_raw |>
  select(all_of(c(trust_items, lapop_demo_source, "pn4", "m1"))) |>
  rename(!!!lapop_trust_names) |>
  mutate(across(all_of(setdiff(lapop_demo_source, "q2")), as_character)) |>
  # rename variables & recode values to english
  mutate(
    # bin age
     age_cat = cut(q2,
```

```r
                  breaks = c(17, 29, 44, 59, Inf),
                  labels = c("18-29", "30-44", "45-59", "60+"),
                  right = TRUE),

    # recode gender to factor with female as reference group
    sex = ifelse(q1tc_r == "Mujer/femenino", "Female", "Male"),
    male = factor(sex),

    # recode education
    Edu = case_when(
      edre == "Ninguna" ~ "none",
      edre == "Primaria incompleta" ~ "primary",
      edre == "Primaria completa" ~ "primary",
      edre ==
        "Secundaria o Educación Media Superior/Bachillerato/Preparatoria/Profesional Técnico incompleta
      edre ==
        "Secundaria o Educación Media Superior/Bachillerato/Preparatoria/Profesional Técnico completa"
      edre ==
        "Universitaria, superior no universitaria o técnico universitario completa" ~ "higher_ed",
      edre ==
        "Universitaria, superior no universitaria o técnico universitario incompleta" ~ "higher_ed",
      TRUE ~ NA_character_
    ),
    Edu = factor(Edu,
                 levels = c("none", "primary", "secondary", "higher_ed")),

    # urban factor
    Urban = factor(ifelse(ur=="Urbano", "urban", "rural")),

    # recode occupation status
    Employment = case_when(
      ocup4a == "Trabajando?" ~ "Employed",
      ocup4a == "No está trabajando en este momento pero tiene trabajo?"
      ~ "Employed",
      ocup4a == "Se dedica a los quehaceres de su hogar?" ~ "house_wife",
      ocup4a == "Es estudiante?" ~ "Student",
      ocup4a ==
        "Está jubilado, pensionado o incapacitado permanentemente para trabajar?" ~ "Retired",
      ocup4a == "Está buscando trabajo activamente?" ~ "Unemployed",
      ocup4a == "No trabaja y no está buscando trabajo?" ~ "Unemployed",
      TRUE ~ NA_character_),
    Employment = factor(Employment),

    # recode outcomes
    satis_demo_lapop = pn4,
    prez_rating = m1
    ) |>
  # listwise deletion
  drop_na() |>
  # drop variables we no longer need
  select(!all_of(lapop_demo_source), -sex, -pn4, -m1)

lapop |> glimpse()
```

```
Rows: 1,430
Columns: 20
$ justice_system     <dbl> 7, 5, 5, 1, 2, 4, 1, 3, 1, 4, 2, 2, 3, 6, 1, 1, 6, ~
$ electoral_tribunal <dbl> 6, 4, 1, 3, 2, 4, 5, 5, 4, 7, 1, 4, 5, 6, 1, 4, 7, ~
$ armed_forces       <dbl> 7, 6, 3, 6, 6, 7, 6, 3, 1, 7, 4, 4, 5, 4, 4, 5, 7, ~
$ legislature        <dbl> 7, 5, 2, 6, 4, 7, 3, 1, 1, 6, 2, 4, 1, 5, 1, 2, 6, ~
$ public_ministry    <dbl> 6, 5, 4, 4, 4, 4, 3, 5, 3, 3, 1, 2, 4, 6, 1, 3, 7, ~
$ police             <dbl> 4, 6, 5, 4, 1, 5, 1, 4, 1, 5, 2, 2, 4, 6, 1, 4, 1, ~
$ auditor            <dbl> 7, 4, 4, 5, 4, 4, 3, 4, 3, 5, 1, 2, 5, 4, 1, 3, 7, ~
$ political_parties  <dbl> 7, 5, 1, 3, 1, 4, 2, 3, 1, 2, 2, 2, 4, 5, 1, 3, 1, ~
$ president          <dbl> 7, 5, 4, 5, 1, 7, 5, 5, 5, 6, 5, 6, 6, 5, 4, 7, 6, ~
$ supreme_court      <dbl> 7, 5, 3, 2, 2, 4, 2, 4, 1, 4, 2, 2, 4, 4, 1, 5, 6, ~
$ municipality       <dbl> 7, 6, 3, 2, 4, 4, 1, 3, 1, 2, 5, 4, 4, 7, 1, 5, 6, ~
$ media              <dbl> 7, 6, 1, 4, 2, 4, 1, 1, 1, 6, 1, 2, 2, 6, 1, 2, 7, ~
$ elections          <dbl> 7, 5, 3, 1, 3, 4, 4, 1, 4, 3, 2, 4, 4, 5, 4, 5, 7, ~
$ age_cat            <fct> 30-44, 30-44, 30-44, 18-29, 60+, 30-44, 45-59, 18-2~
$ male               <fct> Female, Female, Female, Male, Male, Male, Male, Fem~
$ Edu                <fct> primary, primary, secondary, secondary, primary, pr~
$ Urban              <fct> rural, rural, rural, urban, rural, rural, urban, ru~
$ Employment         <fct> house_wife, house_wife, Employed, Employed, Employe~
$ satis_demo_lapop   <fct> 1, 3, 2, 3, 2, 2, 2, 2, 3, 2, 1, 2, 2, 2, 3, 2, 2, ~
$ prez_rating        <fct> 1, 3, 2, 2, 2, 2, 2, 2, 2, 3, 2, 2, 2, 2, 3, 2, 1, ~
```

```r
map(1:ncol(lapop), function(C) {
  col_name <- names(lapop)[C]
  lapop |> select(all_of(C)) |> rename(Var=1) |> count(Var) |> mutate(column = col_name, .before = 1) |
    kable()
})
```

```
[[1]]


|column          | Var|   n|
|:---------------|---:|---:|
|justice_system  |   1| 141|
|justice_system  |   2| 146|
|justice_system  |   3| 235|
|justice_system  |   4| 298|
|justice_system  |   5| 302|
|justice_system  |   6| 172|
|justice_system  |   7| 136|

[[2]]


|column              | Var|   n|
|:-------------------|---:|---:|
|electoral_tribunal  |   1| 139|
|electoral_tribunal  |   2| 124|
|electoral_tribunal  |   3| 160|
|electoral_tribunal  |   4| 242|
|electoral_tribunal  |   5| 278|
|electoral_tribunal  |   6| 245|
|electoral_tribunal  |   7| 242|
```

[[3]]

|column       | Var|   n|
|:------------|---:|---:|
|armed_forces |   1| 105|
|armed_forces |   2|  66|
|armed_forces |   3| 128|
|armed_forces |   4| 178|
|armed_forces |   5| 290|
|armed_forces |   6| 295|
|armed_forces |   7| 368|

[[4]]

|column      | Var|   n|
|:-----------|---:|---:|
|legislature |   1| 139|
|legislature |   2| 108|
|legislature |   3| 172|
|legislature |   4| 290|
|legislature |   5| 350|
|legislature |   6| 226|
|legislature |   7| 145|

[[5]]

|column          | Var|   n|
|:---------------|---:|---:|
|public_ministry |   1| 141|
|public_ministry |   2| 119|
|public_ministry |   3| 181|
|public_ministry |   4| 279|
|public_ministry |   5| 313|
|public_ministry |   6| 222|
|public_ministry |   7| 175|

[[6]]

|column | Var|   n|
|:------|---:|---:|
|police |   1| 297|
|police |   2| 215|
|police |   3| 227|
|police |   4| 242|
|police |   5| 244|
|police |   6| 111|
|police |   7|  94|

[[7]]

|column   | Var|   n|
|:--------|---:|---:|
|auditor  |   1| 128|
|auditor  |   2| 123|
|auditor  |   3| 209|
|auditor  |   4| 325|
|auditor  |   5| 308|
|auditor  |   6| 206|
|auditor  |   7| 131|

[[8]]

|column            | Var|   n|
|:-----------------|---:|---:|
|political_parties |   1| 345|
|political_parties |   2| 211|
|political_parties |   3| 240|
|political_parties |   4| 268|
|political_parties |   5| 202|
|political_parties |   6|  98|
|political_parties |   7|  66|

[[9]]

|column    | Var|   n|
|:---------|---:|---:|
|president |   1| 140|
|president |   2|  69|
|president |   3|  81|
|president |   4| 152|
|president |   5| 262|
|president |   6| 254|
|president |   7| 472|

[[10]]

|column        | Var|   n|
|:-------------|---:|---:|
|supreme_court |   1| 148|
|supreme_court |   2| 114|
|supreme_court |   3| 182|
|supreme_court |   4| 251|
|supreme_court |   5| 324|
|supreme_court |   6| 240|
|supreme_court |   7| 171|

[[11]]

|column       | Var|   n|

```
|:------------|---:|---:|
|municipality |   1| 127|
|municipality |   2| 115|
|municipality |   3| 196|
|municipality |   4| 292|
|municipality |   5| 355|
|municipality |   6| 183|
|municipality |   7| 162|

[[12]]


|column | Var|   n|
|:------|---:|---:|
|media  |   1| 177|
|media  |   2| 177|
|media  |   3| 224|
|media  |   4| 278|
|media  |   5| 265|
|media  |   6| 159|
|media  |   7| 150|

[[13]]


|column    | Var|   n|
|:---------|---:|---:|
|elections |   1| 154|
|elections |   2| 131|
|elections |   3| 214|
|elections |   4| 277|
|elections |   5| 302|
|elections |   6| 224|
|elections |   7| 128|

[[14]]


|column  |Var   |   n|
|:-------|:-----|---:|
|age_cat |18-29 | 468|
|age_cat |30-44 | 361|
|age_cat |45-59 | 369|
|age_cat |60+   | 232|

[[15]]


|column |Var    |   n|
|:------|:------|---:|
|male   |Female | 730|
|male   |Male   | 700|

[[16]]
```

|column |Var       |   n|
|:------|:---------|---:|
|Edu    |none      |  25|
|Edu    |primary   | 273|
|Edu    |secondary | 842|
|Edu    |higher_ed | 290|

[[17]]

|column |Var   |    n|
|:------|:-----|----:|
|Urban  |rural |  277|
|Urban  |urban | 1153|

[[18]]

|column     |Var        |   n|
|:----------|:----------|---:|
|Employment |Employed   | 766|
|Employment |house_wife | 377|
|Employment |Retired    |  87|
|Employment |Student    |  88|
|Employment |Unemployed | 112|

[[19]]

|column           |Var |   n|
|:----------------|:---|---:|
|satis_demo_lapop |1   |  88|
|satis_demo_lapop |2   | 765|
|satis_demo_lapop |3   | 475|
|satis_demo_lapop |4   | 102|

[[20]]

|column      |Var |   n|
|:-----------|:---|---:|
|prez_rating |1   | 342|
|prez_rating |2   | 692|
|prez_rating |3   | 295|
|prez_rating |4   |  63|
|prez_rating |5   |  38|

**We are left with n=1,430**

# Latinobarometro Colombia 2023

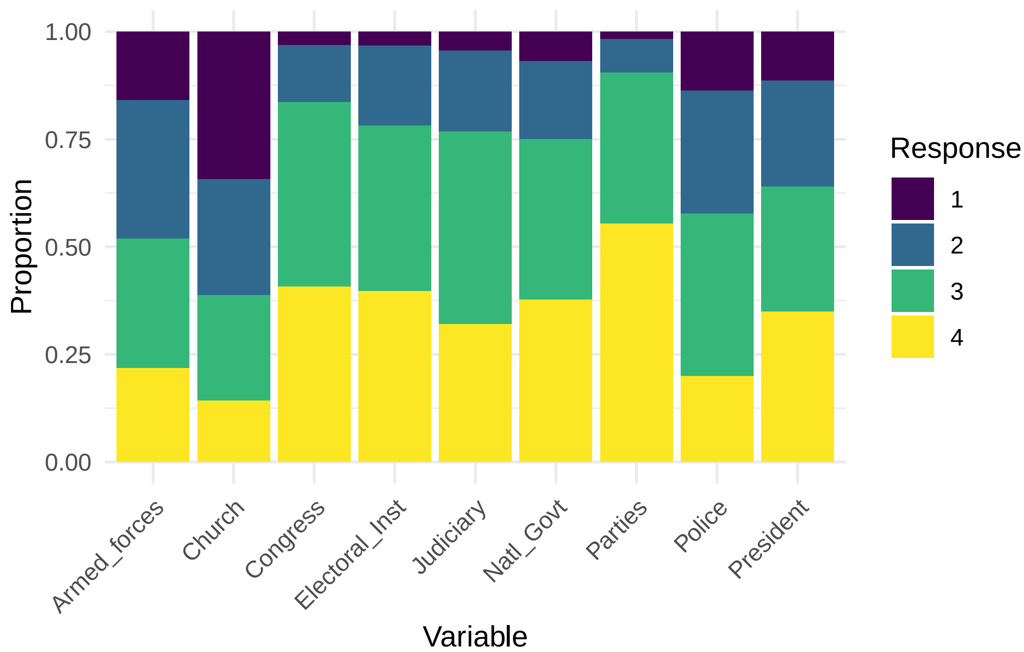
## Load data

```
lb_raw <- read_rds(
  "D:/repos/UMD_classes_code/TSE_II_SURV721/latinobarrometer/Latinobarometro_2023_Esp_Rds_v1_0.rds") |>
  tibble() |>
  # keep only Colombia
  filter(idenpa == 170)
```

#### Trust Battery (9 items) and responses translated to English.

- -5 through -1 = NA
- 1 = A lot of trust
- 2 = some trust
- 3 = little trust
- 4 = no trust

```
trust_vars_raw <- c("P13STGBS_A", "P13STGBS_B", "P13ST_C",
                    "P13ST_D", "P13ST_E", "P13ST_F",
                    "P13ST_G", "P13ST_H", "P13ST_I")

lb_trust_names <- c(Armed_forces = "P13STGBS_A",
                    Police = "P13STGBS_B",
                    Church = "P13ST_C",
                    Congress = "P13ST_D",
                    Natl_Govt = "P13ST_E",
                    Judiciary = "P13ST_F",
                    Parties = "P13ST_G",
                    Electoral_Inst = "P13ST_H",
                    President = "P13ST_I")


lb_raw |>
  select(all_of(trust_vars_raw)) |>
  rename(!!!lb_trust_names) |>
  mutate(across(everything(), ~ifelse(. %in% -5:-1, NA, .))) |>
  pivot_longer(everything(), names_to = "variable", values_to = "value") |>
  drop_na(value) |>
  count(variable, value) |>
  group_by(variable) |>
  mutate(prop = n / sum(n)) |>
  ggplot(aes(x = variable, y = prop, fill = factor(value))) +
  geom_col() +
  scale_fill_viridis_d() +
  labs(x = "Variable", y = "Proportion", fill = "Response") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```




**Outcome Variables**

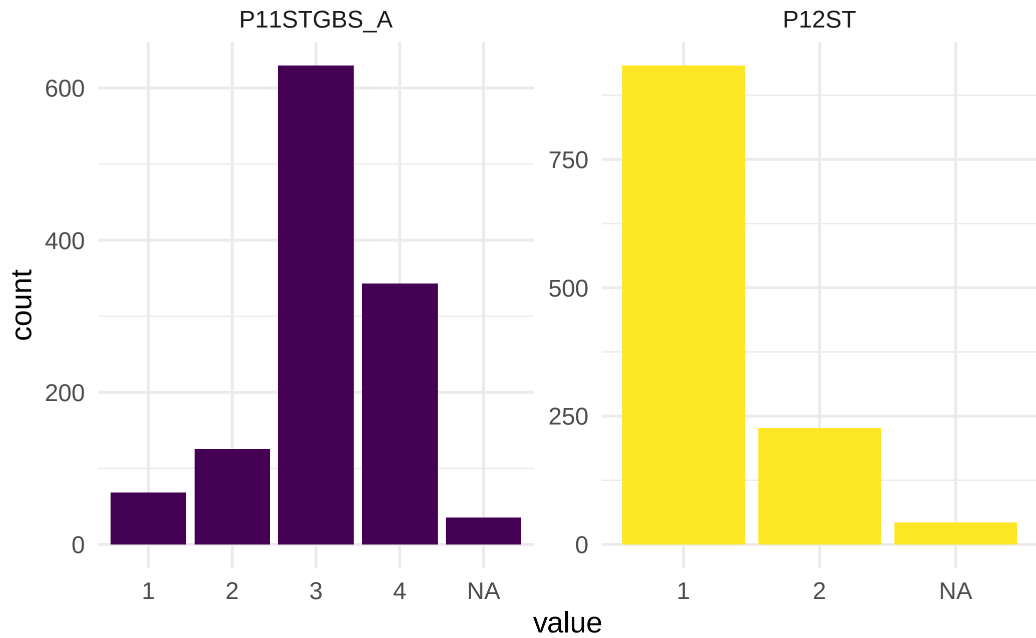
**P11STGBS_A Satisfaction with Democracy codings translated into English.**

- -5 through -1 = NA
- 1 = Very Satisfied
- 2 = Quite Satisfied
- 3 = Not very satisfied
- 4 = not at all satisfied

**P12ST country governed by a few powerful groups, or governed for the good at all codings translated into English**

- -5 through -1 = NA
- 1 = powerful groups in their interest
- 2= for the good of all

```
lb_raw |>
  select(P11STGBS_A, P12ST) |>
  mutate(across(everything(), ~if_else(. < 0, NA_real_, .))) |>
  mutate(across(everything(), factor)) |>
  pivot_longer(everything(), names_to = "variable", values_to = "value") |>
  ggplot(aes(x = value)) +
  geom_bar(aes(fill = variable), show.legend = FALSE) +
  facet_wrap(~variable, scales = "free") +
  scale_fill_viridis_d() +
  theme_minimal()
```




### Recode data

```
# demographics
lb_demo_source <- c("sexo", "edad",
                     "REEEDUC_1", "tamciud", "S18_A")

lb <- lb_raw |>
  select(all_of(trust_vars_raw), all_of(lb_demo_source), P11STGBS_A, P12ST) |>
  rename(!!!lb_trust_names) |>
  mutate(across(all_of(setdiff(lb_demo_source, "edad")), as_character)) |>

  # remove missing data
  mutate(across(everything(), ~ifelse(. %in% -5:-1, NA, .)),

         # recode demographics
          male = factor(
            ifelse(sexo == "Hombre", "Male", "Female")),

         # bin age
         age_cat = cut(edad,
                  breaks = c(17, 29, 44, 59, Inf),
                  labels = c("18-29", "30-44", "45-59", "60+"),
                  right = TRUE),

         # recode education
    Edu = case_when(
      REEEDUC_1 == "Analfabeto" ~ "none",
      REEEDUC_1 == "Basica incompleta" ~ "primary",
```

```
      REEEDUC_1 == "Basica completa" ~ "primary",
      REEEDUC_1 =="Secundaria, media, tecnica completa" ~ "secondary",
      REEEDUC_1 == "Secundaria, media, tecnica incompleta" ~ "secondary",
      REEEDUC_1 == "Superior completa" ~ "higher_ed",
      REEEDUC_1 == "Superior Incompleta" ~ "higher_ed",
      TRUE ~ NA_character_
    ),
    Edu = factor(Edu,
                 levels = c("none", "primary", "secondary", "higher_ed")),

    # urban factor
    Urban = case_when(
      tamciud == "(Capital)" ~ "urban",
      tamciud == "100.001 y mas" ~ "urban",
      tamciud == "50.001 - 100.000" ~ "urban",
      tamciud == "40.001 - 50.000" ~ "urban",
      tamciud == "20.001 - 40.000" ~ "rural",
      tamciud == "10.001 - 20.000" ~ "rural",
      tamciud == "5.001 - 10.000" ~ "rural",
      tamciud == "Menos de 5.000" ~ "rural",
      TRUE ~ NA_character_
    ),

    Urban = factor(Urban, levels = c("rural", "urban")),

    # recode occupation status
    Employment = case_when(
      S18_A == "Independiente/cuenta propia" ~ "Employed",
      S18_A == "Asalariado en emp. privada" ~ "Employed",
      S18_A == "Asalariado en emp. publica" ~ "Employed",
      S18_A ==
        "No trabaja/ responsable de las compras y el cuidado de la casa" ~
        "house_wife",
      S18_A == "Estudiante" ~ "Student",
      S18_A ==  "Retirado/pensionado" ~ "Retired",
      S18_A == "Temporalmente no trabaja" ~ "Unemployed",
      TRUE ~ NA_character_),
    Employment = factor(Employment),

    # recode outcomes
    satis_demo_lb = P11STGBS_A,
    governed_for = P12ST
    ) |>

  # listwise deletion
  drop_na() |>

  # drop variables we no longer need
  select(!all_of(lb_demo_source), -P11STGBS_A, -P12ST)

lb |> glimpse()
```

```
Rows: 1,073
Columns: 16
```

```
$ Armed_forces   <dbl> 4, 4, 1, 2, 3, 1, 3, 2, 3, 3, 1, 3, 2, 2, 3, 2, 2, 4, 2~
$ Police         <dbl> 4, 4, 1, 3, 2, 1, 3, 3, 3, 3, 1, 3, 1, 2, 3, 3, 2, 3, 3~
$ Church         <dbl> 3, 4, 2, 4, 2, 1, 3, 3, 1, 4, 3, 3, 1, 1, 3, 1, 3, 3, 3~
$ Congress       <dbl> 4, 3, 3, 4, 3, 1, 4, 3, 4, 4, 4, 3, 1, 4, 4, 3, 3, 3, 1~
$ Natl_Govt      <dbl> 4, 3, 4, 4, 2, 1, 4, 4, 1, 4, 4, 4, 1, 3, 2, 3, 3, 3, 2~
$ Judiciary      <dbl> 4, 4, 3, 4, 2, 1, 4, 4, 4, 4, 2, 3, 1, 2, 4, 3, 3, 3, 2~
$ Parties        <dbl> 4, 4, 3, 4, 3, 1, 4, 4, 4, 3, 4, 3, 3, 3, 4, 3, 3, 4, 3~
$ Electoral_Inst <dbl> 4, 4, 3, 4, 3, 1, 3, 3, 4, 4, 4, 3, 1, 2, 3, 4, 3, 4, 2~
$ President      <dbl> 4, 4, 3, 4, 2, 1, 2, 4, 3, 4, 4, 4, 1, 3, 3, 3, 2, 4, 2~
$ male           <fct> Female, Male, Female, Male, Female, Male, Male, Female,~
$ age_cat        <fct> 18-29, 30-44, 18-29, 45-59, 45-59, 18-29, 60+, 18-29, 3~
$ Edu            <fct> secondary, higher_ed, secondary, higher_ed, higher_ed, ~
$ Urban          <fct> urban, urban, urban, urban, urban, urban, urban, urban,~
$ Employment     <fct> house_wife, Employed, Student, Employed, house_wife, Em~
$ satis_demo_lb  <dbl> 3, 4, 4, 4, 3, 3, 3, 3, 2, 4, 4, 4, 2, 1, 3, 4, 3, 3, 3~
$ governed_for   <dbl> 1, 1, 1, 1, 1, 2, 1, 1, 2, 1, 1, 1, 2, 1, 1, 1, 1, 1, 2~
```

```
map(1:ncol(lb), function(C) {
  col_name <- names(lb)[C]
  lb |> select(all_of(C)) |> rename(Var=1) |> count(Var) |> mutate(column = col_name, .before = 1) |>
    kable()
})
```

```
[[1]]


|column        | Var|   n|
|:-------------|---:|---:|
|Armed_forces  |   1| 165|
|Armed_forces  |   2| 359|
|Armed_forces  |   3| 320|
|Armed_forces  |   4| 229|

[[2]]


|column | Var|   n|
|:------|---:|---:|
|Police |   1| 142|
|Police |   2| 317|
|Police |   3| 401|
|Police |   4| 213|

[[3]]


|column | Var|   n|
|:------|---:|---:|
|Church |   1| 343|
|Church |   2| 294|
|Church |   3| 276|
|Church |   4| 160|

[[4]]
```

|column | Var| n|
|:--------|---:|---:|
|Congress | 1| 31|
|Congress | 2| 140|
|Congress | 3| 460|
|Congress | 4| 442|

[[5]]

|column | Var| n|
|:---------|---:|---:|
|Natl_Govt | 1| 72|
|Natl_Govt | 2| 198|
|Natl_Govt | 3| 402|
|Natl_Govt | 4| 401|

[[6]]

|column | Var| n|
|:---------|---:|---:|
|Judiciary | 1| 47|
|Judiciary | 2| 208|
|Judiciary | 3| 473|
|Judiciary | 4| 345|

[[7]]

|column | Var| n|
|:-------|---:|---:|
|Parties | 1| 16|
|Parties | 2| 79|
|Parties | 3| 382|
|Parties | 4| 596|

[[8]]

|column | Var| n|
|:--------------|---:|---:|
|Electoral_Inst | 1| 30|
|Electoral_Inst | 2| 203|
|Electoral_Inst | 3| 426|
|Electoral_Inst | 4| 414|

[[9]]

|column | Var| n|
|:---------|---:|---:|
|President | 1| 122|

```
|President |   2| 267|
|President |   3| 308|
|President |   4| 376|

[[10]]


|column |Var    |   n|
|:------|:------|---:|
|male   |Female | 556|
|male   |Male   | 517|

[[11]]


|column  |Var   |   n|
|:-------|:-----|---:|
|age_cat |18-29 | 339|
|age_cat |30-44 | 356|
|age_cat |45-59 | 248|
|age_cat |60+   | 130|

[[12]]


|column |Var       |   n|
|:------|:---------|---:|
|Edu    |none      |  24|
|Edu    |primary   | 153|
|Edu    |secondary | 699|
|Edu    |higher_ed | 197|

[[13]]


|column |Var   |   n|
|:------|:-----|---:|
|Urban  |rural | 368|
|Urban  |urban | 705|

[[14]]


|column     |Var        |   n|
|:----------|:----------|---:|
|Employment |Employed   | 561|
|Employment |house_wife | 263|
|Employment |Retired    |  31|
|Employment |Student    |  86|
|Employment |Unemployed | 132|

[[15]]
```

|column        | Var|   n|
|:-------------|---:|---:|
|satis_demo_lb |   1|  63|
|satis_demo_lb |   2| 111|
|satis_demo_lb |   3| 588|
|satis_demo_lb |   4| 311|

[[16]]

|column       | Var|   n|
|:------------|---:|---:|
|governed_for |   1| 872|
|governed_for |   2| 201|

# We are left with n=1,073

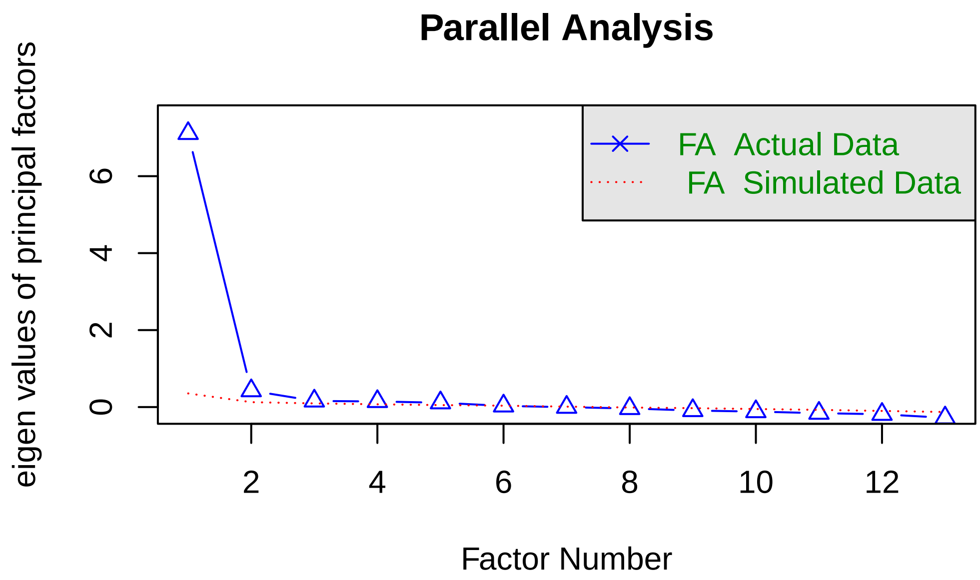
### We now move to exploratory factor analysis for each survey trust items.

#### Lapop EFA with one-three factors

```
#
# Exploratory Factor Analysis - Reusable Function
#

run_efa <- function(items_df, max_factors = 3, seed = 721) {

  # Coerce to numeric, drop incomplete cases
  items_df <- items_df |> drop_na() |> mutate(across(everything(), as.numeric))
  cat(sprintf("EFA: %d cases, %d items\n\n", nrow(items_df), ncol(items_df)))

  # Polychoric correlations - appropriate for ordinal items
  R <- polychoric(items_df)$rho

  # Parallel analysis - compare observed eigenvalues to simulated null
  set.seed(seed)
  pa <- fa.parallel(R, n.obs = nrow(items_df), fa = "fa", cor = "cor",
                    plot = TRUE, main = "Parallel Analysis")

  # Fit 1 through max_factors - oblimin rotation for correlated factors
  models <- map(1:max_factors, \(nf) {
    fa(R, nfactors = nf, n.obs = nrow(items_df),
       rotate = if (nf == 1) "none" else "oblimin",
       fm = "wls", cor = "cor")
  }) |> set_names(paste0("f", 1:max_factors))

  # Comparison table - TLI, RMSEA, BIC for model selection
  comparison <- imap_dfr(models, \(mod, name) {
    tibble(model = name, factors = mod$factors,
           TLI = mod$TLI, RMSEA = mod$RMSEA[1],
           BIC = mod$BIC,
           prop_var = sum(mod$Vaccounted["Proportion Var", ]))
  })
  cat("\n"); print(comparison, width = Inf)

  # Print sorted loadings (cutoff = 0.3) for each solution
  walk(models, \(mod) {
    cat(sprintf("\n-- %d-factor loadings --\n", mod$factors))
    print(mod$loadings, cutoff = 0.3, sort = TRUE)
  })

  # Return key objects for downstream CFA specification
  invisible(list(polychoric = R, parallel = pa, models = models,
                 comparison = comparison, n_suggested = pa$nfact))
}
```

```
lapop_trust <- run_efa(lapop |> select(justice_system:elections), max_factors = 3)
```

```
EFA: 1430 cases, 13 items
```




```
Parallel analysis suggests that the number of factors =  5  and the number of components =  NA

Loading required namespace: GPArotation

Warning in fa.stats(r = r, f = f, phi = phi, n.obs = n.obs, np.obs = np.obs, :
The estimated weights for the factor scores are probably incorrect.  Try a
different factor score estimation method.


# A tibble: 3 x 6
  model factors   TLI  RMSEA   BIC prop_var
  <chr>   <int> <dbl>  <dbl> <dbl>    <dbl>
1 f1          1 0.885 0.113   786.    0.547
2 f2          2 0.908 0.101   447.    0.590
3 f3          3 0.913 0.0985  320.    0.644

-- 1-factor loadings --

Loadings:
                   WLS1
justice_system     0.859
electoral_tribunal 0.720
armed_forces       0.604
legislature        0.778
public_ministry    0.814
police             0.726
auditor            0.829
political_parties  0.771
president          0.542
supreme_court      0.801
municipality       0.736
media              0.579
elections          0.774
```

```
                WLS1
SS loadings    7.110
Proportion Var 0.547

-- 2-factor loadings --

Loadings:
                   WLS1   WLS2
justice_system      0.855
electoral_tribunal  0.803
armed_forces        0.534
legislature         0.733
public_ministry     0.855
police              0.675
auditor             0.861
political_parties   0.729
supreme_court       0.774
municipality        0.721
media               0.647
elections           0.747
president           0.431  0.472

                WLS1  WLS2
SS loadings    6.939 0.582
Proportion Var 0.534 0.045
Cumulative Var 0.534 0.578

-- 3-factor loadings --

Loadings:
                   WLS1   WLS3   WLS2
justice_system      0.785
electoral_tribunal  0.866
public_ministry     0.862
auditor             0.803
political_parties   0.559
supreme_court       0.653
municipality        0.589
media               0.701
elections           0.609
armed_forces               0.802
legislature                0.621
president                         0.990
police              0.372  0.391

                WLS1  WLS3  WLS2
SS loadings    4.929 1.246 1.123
Proportion Var 0.379 0.096 0.086
Cumulative Var 0.379 0.475 0.561
```

- We find that a one factor solution is feasible but has a poor fit indicated by TLI and RMSEA. It seems that trust in the president might be a problematic item.

**We now conduct a CFA one and two factor model on the lapop trust items**

```
#
# CFA - 1-Factor and 2-Factor Models (LAPOP Mexico Trust Items)
#

set.seed(721)

diagnose_cfa <- function(fit, label) {

  cat(sprintf("\n%s\n  %s\n%s\n", strrep("=", 60), label, strrep("=", 60)))

  fm <- fitmeasures(fit, c("chisq.scaled", "df.scaled", "pvalue.scaled",
                            "cfi.scaled", "tli.scaled",
                            "rmsea.scaled", "rmsea.ci.lower.scaled",
                            "rmsea.ci.upper.scaled", "srmr"))
  cat(sprintf("\nchi2 = %.1f (df = %.0f, p = %.4f)\n", fm[1], fm[2], fm[3]))
  cat(sprintf("CFI = %.3f | TLI = %.3f | RMSEA = %.3f [%.3f, %.3f] | SRMR = %.3f\n",
              fm[4], fm[5], fm[6], fm[7], fm[8], fm[9]))

  std <- standardizedSolution(fit) |>
    filter(op == "=~") |>
    select(factor = lhs, item = rhs, loading = est.std, p = pvalue) |>
    filter(!is.na(loading)) |>
    arrange(factor, desc(abs(loading)))
  cat("\nLoadings:\n"); print(as.data.frame(std))

  fcor <- standardizedSolution(fit) |>
    filter(op == "~~", lhs != rhs, lhs %in% std$factor, rhs %in% std$factor)
  if (nrow(fcor) > 0)
    cat(sprintf("\nFactor correlation: %.3f\n", fcor$est.std[1]))

  omega <- tryCatch(compRelSEM(fit), error = function(e) NULL)
  ave   <- tryCatch(AVE(fit), error = function(e) NULL)
  if (!is.null(omega)) cat(sprintf("omega: %s\n",
    paste(names(omega), round(omega, 3), sep = " = ", collapse = ", ")))
  if (!is.null(ave)) cat(sprintf("AVE:   %s\n",
    paste(names(ave), round(ave, 3), sep = " = ", collapse = ", ")))

  mi <- modificationIndices(fit, sort. = TRUE, op = "~~", free.remove = FALSE) |>
    filter(mi > 10) |>
    slice_head(n = 5)
  if (nrow(mi) > 0) {
    cat("\nTop modification indices (> 10):\n")
    print(as.data.frame(mi |> select(lhs, rhs, mi)))
  }

  invisible(fit)
}
```

**We find that the president trust item is problematic and test a 1 factor cfa without.**

```
trust_items_lapop <- c("justice_system", "electoral_tribunal", "armed_forces",
                       "legislature", "public_ministry", "police", "auditor",
```

```r
                     "political_parties", "supreme_court", "municipality",
                     "media", "elections")

trust_lapop_df <- lapop |> select(all_of(trust_items_lapop))

# 1-factor, 12 items, freed residual covariances
mod_1f_trim <- '
  Trust =~ justice_system + electoral_tribunal + armed_forces +
           legislature + public_ministry + police + auditor +
           political_parties + supreme_court + municipality +
           media + elections
  armed_forces ~~ legislature
  public_ministry ~~ auditor
'

fit_1f_trim <- cfa(mod_1f_trim, data = trust_lapop_df,
                   ordered = setdiff(trust_items_lapop, "president"),
                   estimator = "WLSMV")

diagnose_cfa(fit_1f_trim, "1-FACTOR (12 items, 2 freed covariances)")
```

```
============================================================
  1-FACTOR (12 items, 2 freed covariances)
============================================================

chi2 = 698.0 (df = 52, p = 0.0000)
CFI = 0.978 | TLI = 0.972 | RMSEA = 0.093 [0.087, 0.099] | SRMR = 0.033

Loadings:
   factor               item   loading p
1   Trust     justice_system 0.8697208 0
2   Trust            auditor 0.8217229 0
3   Trust    public_ministry 0.8071957 0
4   Trust      supreme_court 0.8060352 0
5   Trust          elections 0.7709621 0
6   Trust  political_parties 0.7684410 0
7   Trust        legislature 0.7627986 0
8   Trust electoral_tribunal 0.7431646 0
9   Trust       municipality 0.7360246 0
10  Trust             police 0.7321006 0
11  Trust              media 0.5967398 0
12  Trust       armed_forces 0.5751952 0
omega: Trust = 0.931
AVE:   Trust = 0.568

Top modification indices (> 10):
                 lhs               rhs       mi
1             police political_parties 33.79123
2      supreme_court         elections 25.89944
3     justice_system     supreme_court 24.59651
4 electoral_tribunal             media 24.39812
5       armed_forces            police 22.79241
```

**12-item unidimensional Institutional Trust factor, with two freed residual covariances (armed_forces ~~ legislature, public_ministry ~~ auditor), president dropped. This is what the data support — the EFA said one dominant factor, the CFA confirms it once the misfit item is removed and substantively justified correlated errors are freed. This will be our production script.**

We move to estimating the correlation between the latent factor and our outcome variables.

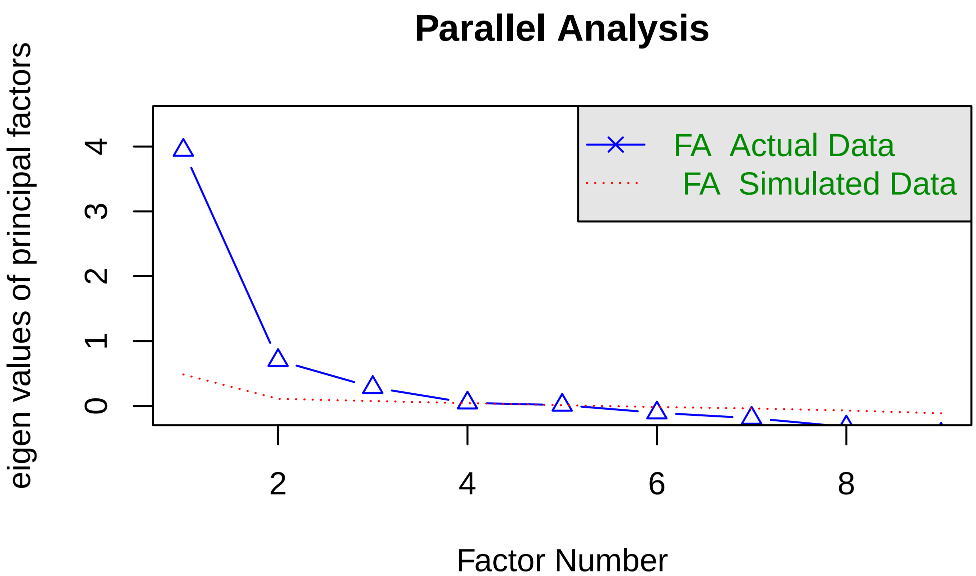
```
# Factor scores
fs <- lavPredict(fit_1f_trim, method = "regression") |> as_tibble()

lapop_fs <- lapop |>
  filter(if_all(all_of(trust_items_lapop), ~ !is.na(.x))) |>
  bind_cols(fs)

# Correlations
cor(lapop_fs$Trust,
    lapop_fs |> select(satis_demo_lapop, prez_rating) |> mutate(across(everything(), as.numeric)),
    use = "complete.obs")
```

```
     satis_demo_lapop prez_rating
[1,]       -0.3326396  -0.2617839
```

We now explore the factor structure of the LB data.

```
lb_trust <- run_efa(lb |> select(Armed_forces:President), max_factors = 3)
```

```
EFA: 1073 cases, 9 items
```




```
Parallel analysis suggests that the number of factors =  5  and the number of components =  NA

# A tibble: 3 x 6
  model factors   TLI RMSEA   BIC prop_var
  <chr>   <int> <dbl> <dbl> <dbl>    <dbl>
1 f1          1 0.636 0.215 1182.    0.439
2 f2          2 0.768 0.172  490.    0.552
3 f3          3 0.693 0.198  431.    0.640

-- 1-factor loadings --

Loadings:
               WLS1
Police         0.544
Congress       0.746
Natl_Govt      0.754
Judiciary      0.779
Parties        0.753
Electoral_Inst 0.764
President      0.664
Armed_forces   0.492
Church

                WLS1
SS loadings    3.947
Proportion Var 0.439
```

```
-- 2-factor loadings --

Loadings:
               WLS1   WLS2
Natl_Govt       0.919
President       0.893
Armed_forces           0.693
Police                 0.771
Church                 0.464
Congress        0.473  0.385
Judiciary       0.495  0.408
Parties         0.499  0.357
Electoral_Inst  0.491  0.389

                WLS1  WLS2
SS loadings    2.609 1.897
Proportion Var 0.290 0.211
Cumulative Var 0.290 0.501

-- 3-factor loadings --

Loadings:
               WLS1   WLS2   WLS3
Congress        0.675
Natl_Govt       0.850
Judiciary       0.651
Parties         0.708
Electoral_Inst  0.661
President       0.802
Armed_forces           0.871
Police                 0.781
Church                        0.851

                WLS1  WLS2  WLS3
SS loadings    3.186 1.421 0.899
Proportion Var 0.354 0.158 0.100
Cumulative Var 0.354 0.512 0.612
```

The Latinobarómetro trust items are clearly not unidimensional. The 1-factor model is unacceptable (TLI = .636, RMSEA = .215), meaning these 9 items cannot be treated as measuring a single construct. The 3-factor solution provides the cleanest simple structure: Factor 1 is political trust (Natl_Govt .85, President .80, Parties .71, Congress .68, Electoral_Inst .66, Judiciary .65), Factor 2 is security institutions (Armed_forces .87, Police .78), and Factor 3 is Church alone (.85). Factor correlations are moderate and distinct (.45, .34, .20), confirming these are separable dimensions. The 2-factor solution is messy — Congress, Judiciary, Parties, and Electoral_Inst all cross-load heavily (complexity ~1.9), meaning the 2-factor structure forces ambiguous assignments. Church is the problem item: it loads .29 on the general factor, forms its own single-item factor in the 3-factor solution, and has near-zero correlations with President (.03) and Natl_Govt (.08). It is substantively distinct from institutional trust and should be dropped. That leaves a 2-factor CFA: Political Trust (6 items) and Security Trust (Armed_forces, Police), with Natl_Govt ~~ President (r = .81 in the polychoric matrix) as a likely candidate for a freed residual covariance.

```
#
# CFA - Latinobarómetro Colombia (8 items, Church dropped)
#
```

```r
set.seed(721)

trust_items_lb <- c("Armed_forces", "Police", "Congress", "Natl_Govt",
                    "Judiciary", "Parties", "Electoral_Inst", "President")

trust_lb_df <- lb |> select(all_of(trust_items_lb))

#   1-factor (baseline)
mod_lb_1f <- '
  Trust =~ Armed_forces + Police + Congress + Natl_Govt +
           Judiciary + Parties + Electoral_Inst + President
'

#   2-factor (EFA-informed)
mod_lb_2f <- '
  Political =~ Congress + Natl_Govt + Judiciary + Parties +
               Electoral_Inst + President
  Security  =~ Armed_forces + Police
'

#   2-factor with freed covariance
mod_lb_2f_free <- '
  Political =~ Congress + Natl_Govt + Judiciary + Parties +
               Electoral_Inst + President
  Security  =~ Armed_forces + Police
  Natl_Govt ~~ President
'

#   Fit all three
fit_lb_1f     <- cfa(mod_lb_1f, data = trust_lb_df,
                      ordered = trust_items_lb, estimator = "WLSMV")
fit_lb_2f     <- cfa(mod_lb_2f, data = trust_lb_df,
                      ordered = trust_items_lb, estimator = "WLSMV")
fit_lb_2f_free <- cfa(mod_lb_2f_free, data = trust_lb_df,
                       ordered = trust_items_lb, estimator = "WLSMV")


#   Diagnose
diagnose_cfa(fit_lb_1f, "LB 1-Factor (8 items)")
```

```
============================================================
  LB 1-Factor (8 items)
============================================================

chi2 = 1360.1 (df = 20, p = 0.0000)
CFI = 0.833 | TLI = 0.766 | RMSEA = 0.250 [0.239, 0.261] | SRMR = 0.124

Loadings:
  factor           item   loading p
1  Trust      Natl_Govt 0.8476955 0
2  Trust      President 0.7826714 0
3  Trust        Parties 0.7418788 0
4  Trust      Judiciary 0.7346768 0
```

```
5  Trust Electoral_Inst 0.7268131 0
6  Trust       Congress 0.7066062 0
7  Trust         Police 0.6405387 0
8  Trust   Armed_forces 0.6179146 0
omega: Trust = 0.95
AVE:   Trust = 0.53

Top modification indices (> 10):
           lhs       rhs        mi
1 Armed_forces    Police 616.87830
2    Natl_Govt President 390.99560
3       Police President 101.02013
4 Armed_forces Natl_Govt  87.19340
5       Police Natl_Govt  81.35119
```

```r
diagnose_cfa(fit_lb_2f, "LB 2-Factor: Political + Security")
```

```
============================================================
  LB 2-Factor: Political + Security
============================================================

chi2 = 600.3 (df = 19, p = 0.0000)
CFI = 0.927 | TLI = 0.893 | RMSEA = 0.169 [0.157, 0.181] | SRMR = 0.076

Loadings:
     factor           item   loading p
1 Political      Natl_Govt 0.8617955 0
2 Political      President 0.8004135 0
3 Political        Parties 0.7593113 0
4 Political      Judiciary 0.7509727 0
5 Political Electoral_Inst 0.7430499 0
6 Political       Congress 0.7228850 0
7  Security         Police 0.8749379 0
8  Security   Armed_forces 0.7869793 0

Factor correlation: 0.519
omega: Political = 0.901, Security = 0.764
AVE:   Political = 0.6, Security = 0.692

Top modification indices (> 10):
        lhs       rhs        mi
1 Natl_Govt President 291.23538
2 Natl_Govt   Parties  40.77779
3   Parties President  37.69564
4  Congress President  33.45876
5  Congress   Parties  32.00420
```

```r
diagnose_cfa(fit_lb_2f_free, "LB 2-Factor + freed Natl_Govt ~~ President")
```

```
============================================================
  LB 2-Factor + freed Natl_Govt ~~ President
============================================================
```

```
chi2 = 113.9 (df = 18, p = 0.0000)
CFI = 0.988 | TLI = 0.981 | RMSEA = 0.071 [0.058, 0.083] | SRMR = 0.033

Loadings:
     factor           item   loading p
1 Political      Judiciary 0.7948272 0
2 Political        Parties 0.7896473 0
3 Political Electoral_Inst 0.7798566 0
4 Political       Congress 0.7597465 0
5 Political      Natl_Govt 0.7064816 0
6 Political      President 0.6030639 0
7  Security         Police 0.8764833 0
8  Security   Armed_forces 0.7855917 0

Factor correlation: 0.550
omega: Political = 0.793, Security = 0.764
AVE:   Political = 0.551, Security = 0.693
```

```r
#   LRT comparison
lavTestLRT(fit_lb_1f, fit_lb_2f)
```

```

Scaled Chi-Squared Difference Test (method = "satorra.2000")

lavaan->lavTestLRT():
   lavaan NOTE: The "Chisq" column contains standard test statistics, not the
   robust test that should be reported per model. A robust difference test is
   a function of two standard (not robust) statistics.

          Df AIC BIC  Chisq Chisq diff   RMSEA Df diff Pr(>Chisq)
fit_lb_2f 19         345.10
fit_lb_1f 20         961.08     274.38 0.75706       1  < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The 2-factor with freed Natl_Govt ~~ President is the Colombia measurement model. CFI = .988, TLI = .981, RMSEA = .071, SRMR = .033 — all criteria pass. The factor correlation at .550 is moderate, confirming Political and Security are genuinely distinct dimensions (unlike the LAPOP 2-factor at .910). The LRT confirms the 2-factor is significantly better than 1-factor.

```r
fs_lb <- lavPredict(fit_lb_2f_free, method = "regression") |> as_tibble()

lb_fs <- lb |>
  filter(if_all(all_of(trust_items_lb), ~ !is.na(.x))) |>
  bind_cols(fs_lb)

cor(lb_fs |> select(Political, Security),
    lb_fs |> select(satis_demo_lb, governed_for) |> mutate(across(everything(), as.numeric)),
    use = "complete.obs")
```

```
          satis_demo_lb governed_for
Political     0.3701401  -0.28595316
Security      0.2124642  -0.06715334
```

**Political Trust → satis_demo_lb (r = .37):** Higher factor score (less trust) → higher satis_demo values (less satisfied). Equivalently, people who trust political institutions more are more satisfied with democracy. Intuitive direction, moderate-strong relationship.

**Political Trust → governed_for (r = -.29):** Higher factor score (less trust) → lower governed_for (1 = powerful groups). People with less political trust think the country is governed by powerful groups. Intuitive direction, moderate relationship.

**Security → satis_demo_lb (r = .21):** Same direction as Political but weaker. Security institution trust matters less for democratic satisfaction.

**Security → governed_for (r = -.07):** Essentially zero. Whether you trust the armed forces and police has almost no bearing on whether you think the country is governed for everyone's benefit. This makes substantive sense — governance perceptions are about political power, not security institutions.

We drop the security factor and find cleaner results.

| | LAPOP Mexico | Latinobarómetro Colombia |
|---|---|---|
| **Items** | 12 (of 13, president dropped) | 6 (of 9, Church/Armed_forces/Police dropped) |
| **Factors** | 1 (Trust) | 1 (Political) |
| **Freed covariances** | armed_forces ~~ legislature, public_ministry ~~ auditor | Natl_Govt ~~ President |
| **CFI / TLI** | .964 / .955 | .992 / .985 |
| **RMSEA** | .056 | .077 |
| **omega** | .922 | .793 |
| **AVE** | .521 | .551 |
| **→ satis_demo** | r = -.33 | r = .37 |
| **→ prez_rating / governed_for** | r = -.26 | r = -.30 |

```
#
# Final CFA Models - Both Surveys
#

#   LAPOP Mexico: 12-item unidimensional Trust
# President dropped (weak loading, single-item EFA factor, misfit source)
# Two freed residual covariances (substantively justified):
#   armed_forces ~~ legislature    (state power institutions)
#   public_ministry ~~ auditor     (oversight/accountability bodies)

lapop_cfa <- '
  Trust =~ justice_system + electoral_tribunal + armed_forces +
           legislature + public_ministry + police + auditor +
           political_parties + supreme_court + municipality +
           media + elections
  armed_forces ~~ legislature
  public_ministry ~~ auditor
'

#   Latinobarómetro Colombia: 6-item unidimensional Political Trust
# Church dropped (h2 = .085, near-zero correlations with political items)
# Armed_forces and Police dropped (Security factor uncorrelated with outcomes)
# One freed residual covariance (substantively justified):
#   Natl_Govt ~~ President  (polychoric r = .81, conceptual overlap)

lb_cfa <- '
  Political =~ Congress + Natl_Govt + Judiciary + Parties +
               Electoral_Inst + President
  Natl_Govt ~~ President
'
```

### To save data into a .rds we will combine both surveys into a list wtih two elements.

```
survey_list <- list(
  # drop president variable
  lapop |> select(-president),
```

```
  # drop armed forces, police, and church
  lb |> select(-Armed_forces:-Church)) |>
  set_names(c("lapop", "lb"))

write_rds(survey_list,
          file = str_c(
            results_dir, "/survey_data.rds")
          )
```